

Dibyendu Kuiri | Curriculum Vitae

PhD in Physics (2026), ACMiN, AGH University of Krakow – Krakow, Poland

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Ph.D. in Theoretical Condensed Matter Physics from AGH University of Krakow (2026), specializing in quantum transport and topological superconductivity in hybrid semiconductor–superconductor nanostructures. Experienced in theoretical and computational condensed-matter physics employing tight-binding methods, scattering theory, Green's-function techniques, and large-scale simulations using Kwant and Tkwant. Current research directions include open quantum systems, non-Hermitian physics, and altermagnetism as a platform for topological superconductivity.

Education

- **Department of Theory of Quantum Systems, ACMiN, AGH University of Krakow.** **Kraków, Poland**
PhD in Physics **2021–2026**
Dissertation: **Electronic transport and topological superconductivity in nanoscopic Josephson junctions**
Defended: **13 March 2026**
Degree awarded: **19 March 2026**
- **Department of Physics, Savitribai Phule Pune University (University of Pune)** **Pune, India**
Junior Research Fellow (DST-SERB Project) **2018–2021**
Research Topic: Vortex dynamics in Bose–Einstein condensates
- **Department of Physics, Savitribai Phule Pune University (University of Pune)** **Pune, India**
M.Sc. in Physics (Theoretical Condensed Matter) **2016–2018**
- **Department of Physics, Raghunathpur College** **Purulia, India**
B.Sc. in Physics (Honours) **2013–2016**

Publication

- Published (2026): **Impact of spin-orbit coupling and Zeeman interaction on the subharmonic gap structure due to multiple Andreev reflections in nanoscopic Josephson junctions**, (D. Kuiri, J. H. Correa, A. Biborski, M. P. Nowak, [Scientific Reports](#).)
- Published (2025): **Enhancement of the topological regime in elongated Josephson junctions**, (D. Kuiri, P. Wójcik, M. P. Nowak, [Phys. Rev. B 111, 085416 \(2025\)](#).)
- Published (2024): **Controlling vortex lattice structure of binary Bose-Einstein condensates via disorder-induced vortex pinning**, (D. Kuiri, T. Mithun, and B. Dey, [Phys. Rev. A 109, 063333](#).)
- Published (2023): **Non-local transport signatures of topological superconductivity in a phase-biased planar Josephson junction**, (D. Kuiri, M. P. Nowak, [Phys. Rev. B 108, 205405 \(2023\)](#).)
- Published (2022): **Controlling Andreev Bound States with the Magnetic Vector Potential**, (CM Moehle, PK Rout, NA Jainandunsing, D Kuiri, C. T. Ke, D. Xiao, C. Thomas, M. J. Manfra, M. P. Nowak, S. Goswami, [Nano Lett. 22, 8601 \(2022\)](#).)

Research Experience

- **Department of Theory of Quantum Systems, ACMiN, AGH University of Krakow, Poland** **10 October 2021–19 March 2026**
Doctoral Student (PhD student)
Worked on *electronic transport and topological superconductivity in nanoscopic Josephson junctions* under the supervision of [Prof. Michał P. Nowak](#).

- **Department of Physics,
Savitribai Phule Pune University
Junior Research Fellow (DST-SERB Project)**

05 August 2018–22 January 2021

Worked on *vortex dynamics in two-component Bose–Einstein condensates* under the supervision of [Prof. Bishwajyoti Dey](#) and [Dr. Mithun Thudiyangal](#).

- **NCL Pune, India
M.Sc. Project**

July 2017–July 2018

Worked on *density functional theory and its applications to surface reactions* under the supervision of [Dr. Kavita Joshi](#).

Research Interests

- **Non-Hermitian physics and open quantum systems:**
Non-Hermitian terms arise naturally when a quantum system couples to external leads, producing complex eigenvalues, non-orthogonal eigenstates, and exceptional points. I study how spin-selective dissipation and asymmetric lead coupling modify the spectral properties and phase boundaries of topological superconductors, particularly to Majorana-bound states in proximitized nanowires and multiterminal Josephson junctions. Transport signatures, including zero-bias conductance peaks, nonlocal conductance, and subgap spectroscopy, serve as experimental probes of these effects.
- **Quantum transport and topological superconductivity in hybrid systems:**
Analysis of transport in semiconductor–superconductor heterostructures, including multiple Andreev reflections and proximity-induced superconductivity, with emphasis on Andreev bound states (ABSs) and Majorana bound states (MBSs) in nanowires, hybrid quantum dots, and multiterminal Josephson junctions (JJs).
- **Altermagnetism and spin-polarized transport:**
Study of superconductor–altermagnet Josephson junctions, where momentum-dependent spin splitting reduces the external magnetic field required for topological superconductivity. Numerical simulations of ABSs and MBSs using Kwant, with analysis of pairing symmetry and altermagnetic parameters governing the emergence of MBSs. Ongoing interest in transport properties, including local and nonlocal conductance signatures.
- **Vortex dynamics in Bose–Einstein condensates:**
Vortex lattice formation and dynamics in two-component Bose–Einstein condensates, studied via numerical solutions of coupled Gross–Pitaevskii equations.

Technical and Personal Skills

- **Programming:** Python, KWANT, TKWANT, Fortran, L^AT_EX
- **Parallel Computing:** OpenMP, Python parallelism using `ipyparallel` and multiprocessing
- **Software:** Inkscape, GNU Octave/matlab (beginner)
- **Scientific Software:** Origin Statistical Analysis Tool (beginner)
- **System Administration:** Managing Linux servers
- **Code Availability:** All developed codes are openly available on [Zenodo](#)
Example repositories: [\[Zenodo Record 8086328\]](#), [\[Zenodo Record 13740205\]](#), [\[Zenodo Record 17544732\]](#)

Oral Presentations

- **International Physics Student Conference SMOK 2025,**
Kraków, Poland, 25-27 April 2025.
Title: *Topological superconductivity in hybrid Josephson junctions.*
- **Uncovering physics: emerging research and future possibilities,**
Raghunathpur, India, 23-24 December 2024.
Title: *Topological superconductivity in Planar Josephson junctions.*
- **3rd Off-Site Scientific Symposium of Academic Centre for Materials and Nanotechnology of AGH,**
Hotel STASINDA, Bukowina Tatrzańska, October, 16-18, 2024.
Title: *Enhancement of topological regime in elongated Josephson junctions.*
- **52nd International School & Conference on the Physics of Semiconductors "Jaszowiec 2024",**
Szczyrk, Poland, 15-21 June 2024.

Title: *Enhancement of topological phase in long Josephson junctions*

- **2nd Off-Site Scientific Symposium of Academic Centre for Materials and Nanotechnology of AGH,**
Hotel Geovita, Zakopane, October 5-7, 2022.

Title: *Numerical simulations of spectroscopy of Andreev bound states in a Josephson junction.*

Poster Presentations

- **The 26th international conference on Electronic Properties and 2D Systems & the 22nd international conference on Modulated Semiconductor Structures,**
St. Louis, Missouri, USA, 27 July – 1 Aug, 2025.
Title: *Impact of spin-orbit coupling, magnetic field, and band mixing on multiple Andreev reflections in nanoscopic Josephson junctions.*
- **53rd international school & conference on the Physics of semiconductors, "Jaszowiec 2025",**
Szczyrk, Poland, 7-13 June 2025.
Title: *Multiple Andreev reflections in spinfull nanoscopic Josephson junctions.*
- **MesoSchool 2024, 2D materials, superconductivity and superconducting circuits, quantum transport and hybrid systems. Theory and experiments.**
Institute of Scientific Studies of Cargèse, Corsica, France, 30th Sept - Oct 12th, 2024.
Title: *Topological phase enhancement in long Josephson junction.*
- **Quantum Matter International Conference – QUANTUMatter 2024**
07-10 May, San Sebastián, Spain.
Title: *Topological phase enhancement in planar Josephson junction in the long junction regime.*
- **The 25th International Conference on the Electronic Properties of Two-Dimensional Systems (EP2DS-25) and 21st International Conference on Modulated Semiconductor Structures (MSS-21)**
Grenoble, France, 9–14 July 2023.
Title: *Non-local spectroscopy of topological superconductivity in Josephson junctions.*
- **51th International School & Conference on the Physics of Semiconductors, "Jaszowiec 2023",**
Szczyrk, Poland, 17-23 June 2023.
Title: *Non-local Transport Signatures of Topological Superconductivity in Planar Josephson Junctions.*
- **50th International School & Conference on the Physics of Semiconductors "Jaszowiec 2022",**
Szczyrk, Poland, 4-10 June 2022.
Title: *Spectroscopy of Andreev bound states in a hybrid semiconductor-superconductor Josephson junction.*
- **Raman Memorial Conference-2020,**
Department of Physics, Savitribai Phule Pune University, Pune, India.
Title: *Effect of impurities on vortex dynamics in two-component Bose-Einstein Condensate.*

Workshop and School

- **Online School and Discussion Meeting on Trapped Atoms, Molecules and Ions:**
International Centre for Theoretical Science (ICTS) from 10-22 May, 2021, Bengaluru, India
- **Online lecture series on Superconductivity**
Organized by Prof. Pratap Raychaudhuri, TIFR Mumbai (2 Sept- 30 Dec, 2020), India.
- **Online workshop on python computing :**
IISER Kolkata. (12- 24 June 2020), West Bengal, India

Honors and Awards

- **IDUB D1, 2025:**
Additional financing participation in international conferences for young researchers and doctoral students – 2nd edition
- **IDUB D5 Scholarship October, 2024 - September, 2025:**
Awarded by the AGH University for publications in reputed journals and scientific achievements as a doctoral student in the preceding academic year.
- **An increase in the doctoral scholarship financed from the subsidy, October, 2024 - September, 2025.**
- **An increase in the doctoral scholarship financed from the subsidy, October, 2023 - September, 2024.**

- Awarded a one-time scholarship in accordance with the recommendation of the Ph.D. Committee of the Physical Education Fund in 10 Feb, 2023.
- Awarded a one-time scholarship to doctoral students with special publishing activity in 2022, AGH, Krakow, Poland.
- **Sonta Bis scholarship with grant number:2020/38/E/ST3/00418:**
Awarded by the National Science Centre Grant for the four-year doctoral study in AGH University of Krakow.
- Awarded scholarship under DST-SERB project from 2018 to 2021.
- Qualified GATE in Physical Science. GATE score 361
- **Qualified in the CSIR-NET:**
Held in June 2018 for an all-India examination for admission to Ph.D. and eligibility for lectureship in Physical sciences.

Hobbies and Interests:

- Enjoy watching science documentaries.
- Advanced salsa dancer; also enjoy playing badminton and cricket.
- Passionate about traveling, hiking, and cooking.
- Languages: Bengali (native), English (fluent), Hindi, Marathi, Polish (basic), Spanish (basic).